



FOR OFFICIAL USE

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National
Qualifications
2024

Mark

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X823/75/01**Engineering Science**

FRIDAY, 17 MAY

9:00 AM – 10:50 AM



* X 8 2 3 7 5 0 1 *

Fill in these boxes and read what is printed below.

Full name of centre

--

Town

--

Forename(s)

--

Surname

--

Number of seat

--

Date of birth

Day

--	--

Month

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Year

--	--

Scottish candidate number

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Total marks — 110**SECTION 1 — 20 marks**

Attempt ALL questions.

SECTION 2 — 90 marks

Attempt ALL questions.

Show all working and units where appropriate.

You should refer to the National 4/5 Engineering Science Data Booklet which you have been given.

The number of significant figures expressed in a final answer should be equivalent to the least significant data value given in the question. Answers that have two more figures or one less figure than this will be accepted.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



* X 8 2 3 7 5 0 1 0 1 *

SECTION 1 — 20 marks

Attempt ALL questions

1. Digital electronic signals can be controlled using logic gates.

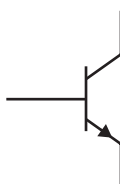
- (a) State the logic gate with a **single** input that will change a high input to a low output.

1

- (b) Draw the symbol for an AND gate.

1

2. The symbol for an electronic component is shown below.



- (a) State the name of this component.

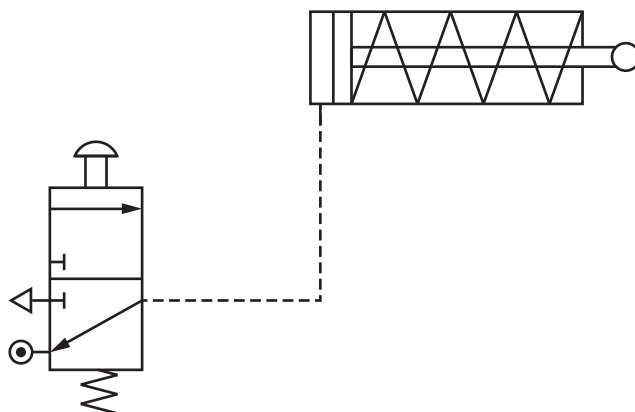
1

- (b) Indicate with an X, the location of the **base** connection on the symbol shown above.

1



3. A diagram for a pneumatic circuit, designed to outstroke a piston when a push button is pressed on a valve, is shown.



Errors were made in the circuit diagram shown above.

- (a) Describe two errors that were made in this circuit diagram.

2

Error 1 _____

Error 2 _____

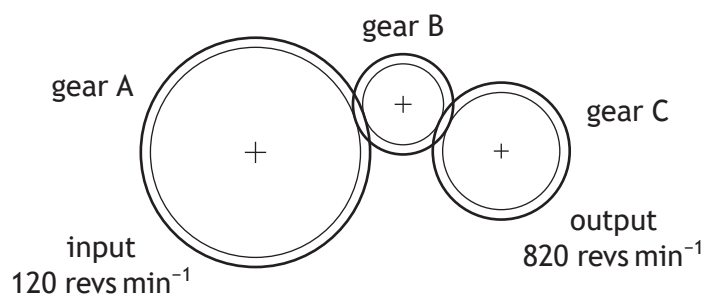
- (b) State the type of engineer that would be responsible for designing a pneumatic circuit.

1

[Turn over



4. A diagram of a simple gear train is shown below.



- (a) Calculate the velocity ratio of this simple gear train.

2

- (b) State the effect on the output velocity if the number of teeth on gear B is reduced.

1



5. A wind farm is being developed on a Scottish island.



- (a) State the type of engineer that would be responsible for monitoring:

(i) the construction of the access roads to the wind turbines

1

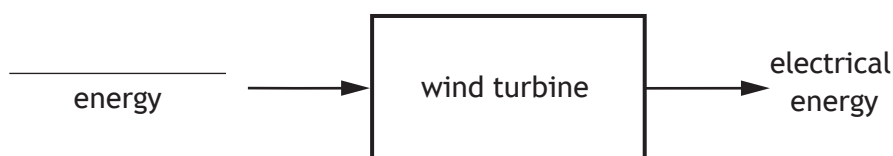
(ii) the impact on the land during installation of the wind turbines.

1

An incomplete diagram showing the **energy** transformation in a wind turbine is shown below.

- (b) Complete the diagram by adding the missing input **energy**.

1



The wind turbine uses feedback in its operation.

- (c) State the type of control that uses feedback.

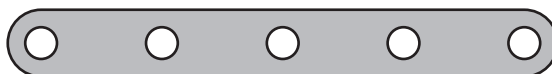
1



6. Members in a frame structure are tested using compressive and tensile forces.

- (a) (i) Draw two arrows which represent an external **compressive** force on member A shown below.

1



member A

- (ii) Draw two arrows which represent an external **tensile** force on member B shown below.

1



member B

Member B has a cross-sectional area of 36 mm^2 .

- (b) Calculate the stress on member B when a 1152 N force is applied.

2



7. Friction can cause the moving parts of a drive system to wear out.

(a) Describe how to reduce the wear on the moving parts of a drive system.

1

(b) State a form of energy lost due to friction in a drive system.

1

[Turn over



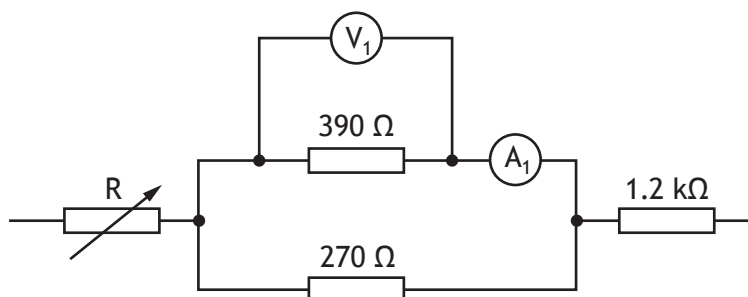
SECTION 2 — 90 marks

Attempt ALL questions

8. A child's microphone with built-in speaker is shown.



Part of the circuit used to control the volume of the speaker is shown.



- (a) For the circuit shown above:

- (i) calculate the resistance of the **parallel** branch

2



8. (a) (continued)

- (ii) calculate the **total resistance** of the circuit when the variable resistor R is set to $780\ \Omega$.

2

- (b) Calculate the current reading on ammeter A_1 when the voltage V_1 is 1.2 V.

3

The resistance of the variable resistor R in the circuit is **increased**.

- (c) State the effect this will have on the:

(i) voltage V_1 _____

1

(ii) current A_1 _____

1

During the development of the child's microphone and speaker, computer simulation was used to test the complete circuit before constructing a prototype.

- (d) Explain why it is an advantage to test the circuit using computer simulation before constructing the prototype.

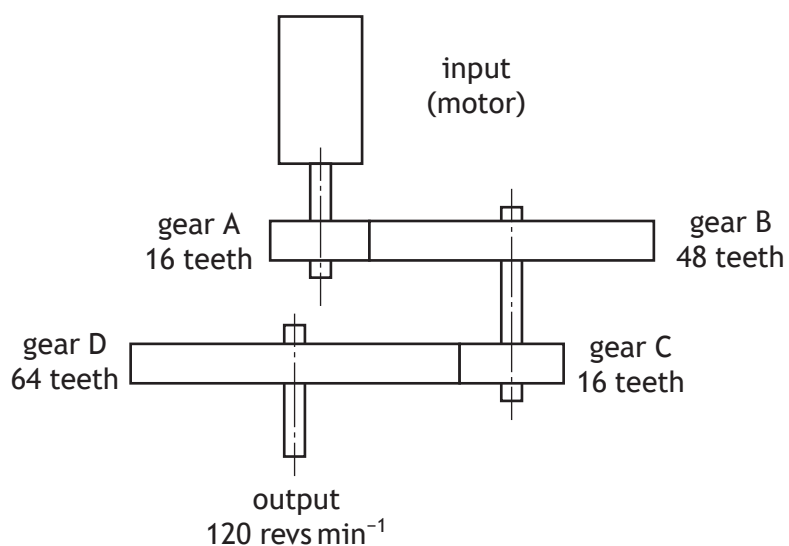
2



9. An electric bike uses a motor to assist a cyclist when pedalling. An electric bike and charging point are shown.



The motor and gear train used in the electric bike are shown below.



9. (continued)

(a) Calculate the rotational speed of the motor.

4

The motor is rated at 36 V and 250 W.

(b) Calculate the current supplied to the motor.

3



9. (continued)

(c) Explain how the efficiency of the gear train could be improved.

2

(d) Describe one positive **environmental** impact of using an electric bike.

1

(e) Describe one positive **social** impact of using an electric bike.

1

(f) Describe one negative **economic** impact of using an electric bike.

1



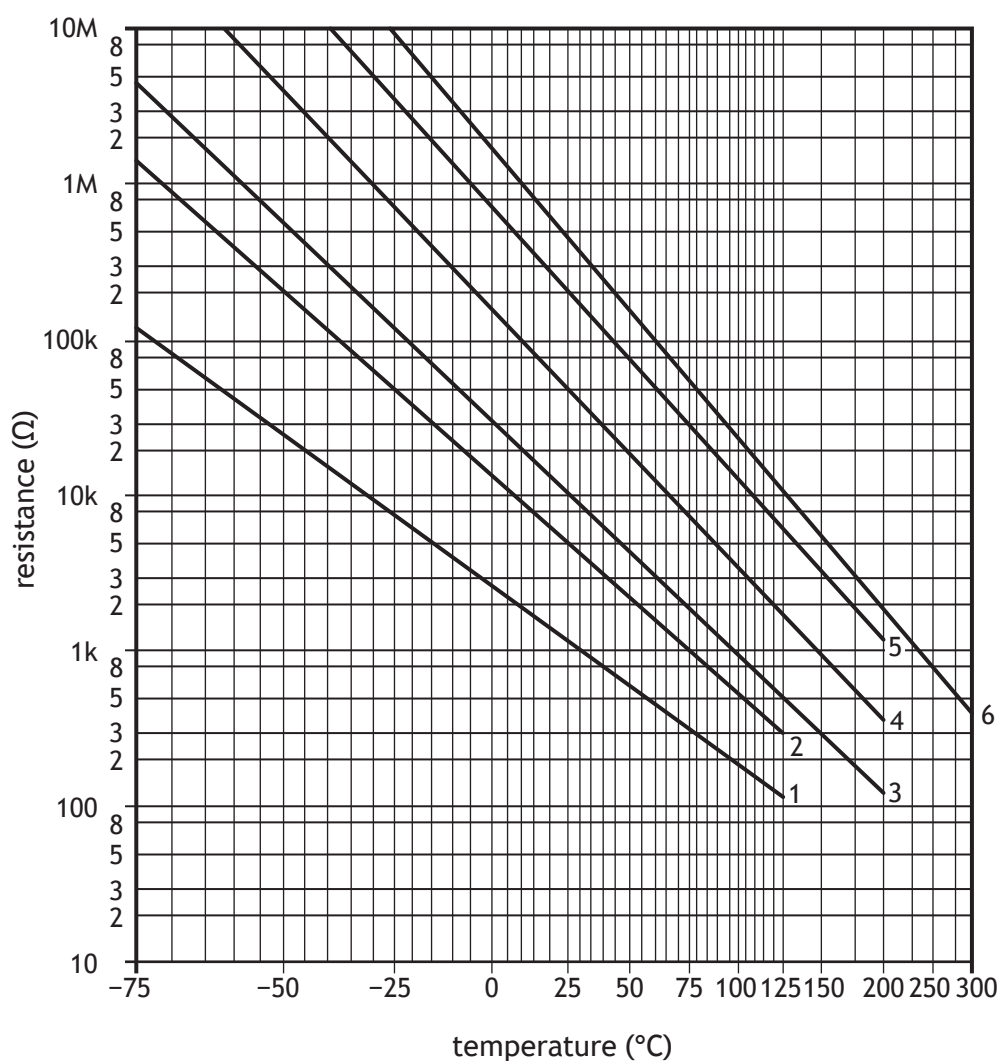
10. A kettle is shown below.

MARKS
DO NOT
WRITE IN
THIS
MARGIN



A thermistor is used as part of a sensor to measure the water temperature in the kettle.

The operating characteristics for a range of thermistors are shown on the graph below.



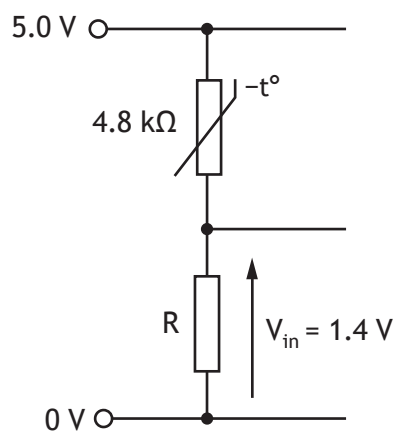
- (a) Determine the resistance of a **type 2** thermistor at a temperature of 85 °C, with reference to the graph.

1



10. (continued)

The sensing circuit used to detect the temperature of the water is shown below.



(b) State the name of this type of series circuit.

1

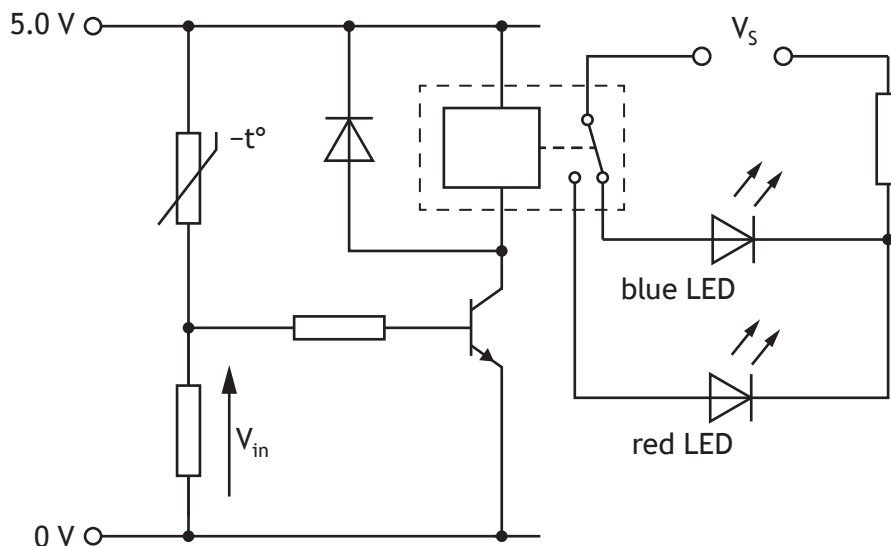
(c) Calculate the resistance R in the sensing circuit shown above.

4



10. (continued)

The kettle has coloured LEDs to indicate the temperature of the water. A simplified circuit used to model the operation of the LEDs is shown below.



- (d) Describe the operation of this circuit as the water temperature increases. Include reference to the resistance of the thermistor, the voltage V_{in} and both LEDs.

5

As the water temperature increases . . . _____

[illegible]

10. (continued)

A kettle uses established technology.

An **emerging technology** is one that is being developed and still to be tried commercially within a product or system.

- (e) Explain a possible impact of an emerging technology which you are familiar with.

2

Name of emerging technology _____

Impact _____



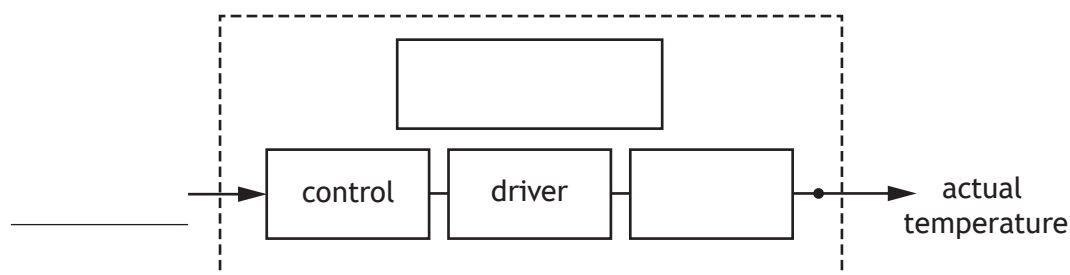
11. A dishwasher in a restaurant kitchen is shown.



Part of the specification for the dishwasher is given below:

- A user selects a set temperature.
- A sensor detects the actual water temperature and sends a signal to the control.
- The control compares the actual water temperature with the set temperature.
- If the actual water temperature is lower than the set temperature a heating element will switch on.

An incomplete sub-system diagram for the dishwasher is shown below.



(a) Complete the sub-system diagram with reference to the specification.

4

[Turn over



11. (continued)

The dishwasher is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	red LED
	6	water pump
	5	heating element
	4	buzzer
	3	
start switch	2	
door sensor	1	
	0	

A drying cycle used in the dishwasher operates using the following sequence:

- When a start switch is pressed a red LED will turn on.
- After a door sensor is activated a water pump will turn on for 35 seconds and then turn off.
- A heating element will then turn on for 5 minutes and then turn off.
- A buzzer will then sound six times, turning on for 0.3 seconds and turning off for 0.3 seconds each time.
- The red LED will then turn off.
- The sequence will return to the start.

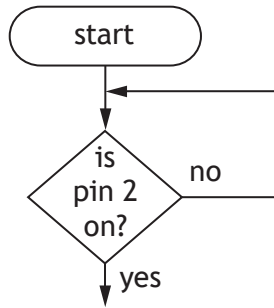
(b) Complete the flowchart shown opposite for the sequence with reference to the data booklet and input/output connections.

Include **all** pin numbers and delay units in your flowchart.

11



11. (b) (continued)



[Turn over



12. A portable power supply used when camping can produce a range of output voltages.

The portable power supply can be charged using solar panels as shown below.



- (a) Explain why using solar panels can contribute towards limiting climate change. 2

A 24 V mini fridge is powered by the portable power supply for 7.5 hours.

- (b) Calculate the electrical energy used by the mini fridge when the current is 0.42 A. 3



12. (continued)

The Boolean equation for part of a circuit used in the portable power supply is shown.

$$Z = (A + B) \bullet (\bar{A} + C)$$

(c) Draw the logic diagram for this Boolean equation.

4

A ○

B ○

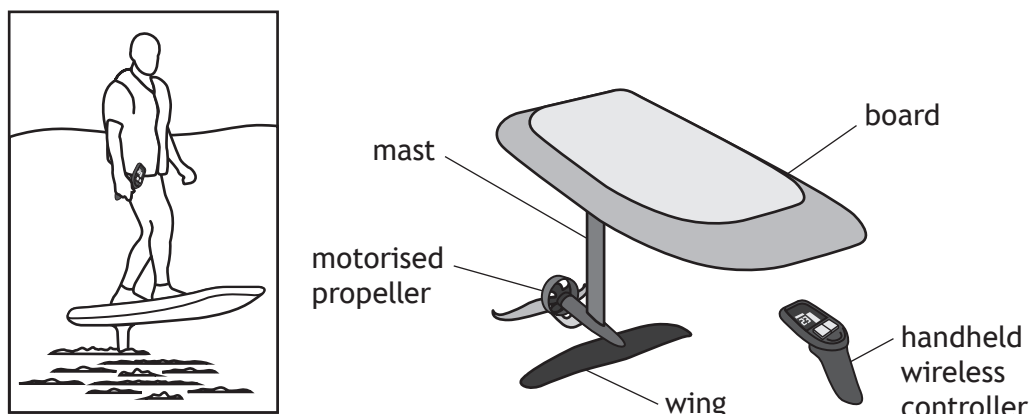
○ Z

C ○

[Turn over



13. A hydrofoil board allows a rider to glide over water.
The hydrofoil board and handheld wireless controller are shown.



Several different engineers were involved during their development.

- (a) Describe a task that each of the following types of engineer would complete during the **development** of the hydrofoil board and handheld wireless controller:

(i) mechanical engineer

1

(ii) structural engineer

1

(iii) electronic engineer.

1



13. (continued)

The hydrofoil board and rider have a combined mass of 115 kg.

- (b) Calculate the velocity of the hydrofoil board and the rider when their kinetic energy is 13 kJ.

3

The hydrofoil board is to be used in the sea and must be able to withstand a range of different forces.

The properties of four materials considered for the mast are shown in the table below.

Material	Corrosion resistant	Strength
A	yes	low
B	no	high
C	yes	high
D	yes	medium

- (c) Select, from the table, the most suitable material to be used for the mast and justify your choice.

2

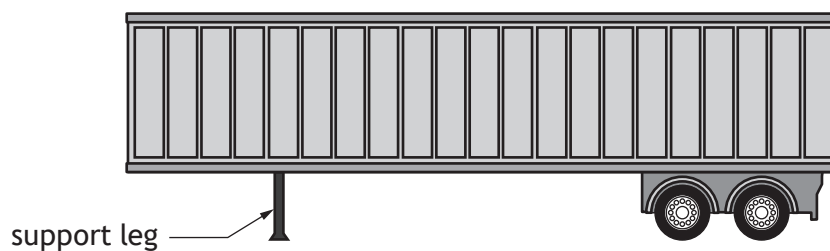
Selected material _____

Justification _____

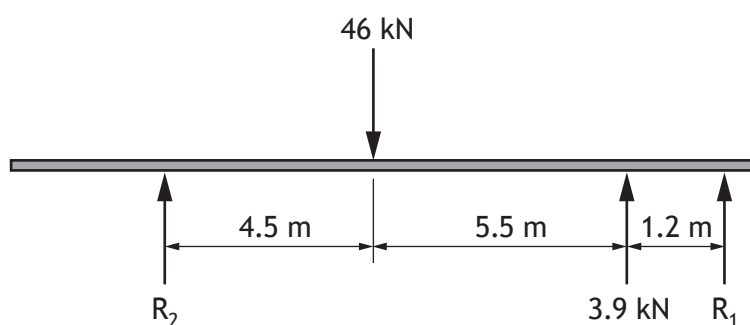
[Turn over



14. A refrigerated lorry trailer used to transport fresh food is shown below.



A simplified free body diagram for the empty lorry trailer is shown.



- (a) (i) Calculate the reaction force R_1 by taking moments about R_2 .

3



14. (a) (continued)

(ii) Calculate the reaction force R_2 .

2

When loaded, the lorry trailer's support leg has a strain of 0.00038 and a change in length of 0.46 mm.

(b) Calculate the original length of the support leg.

3

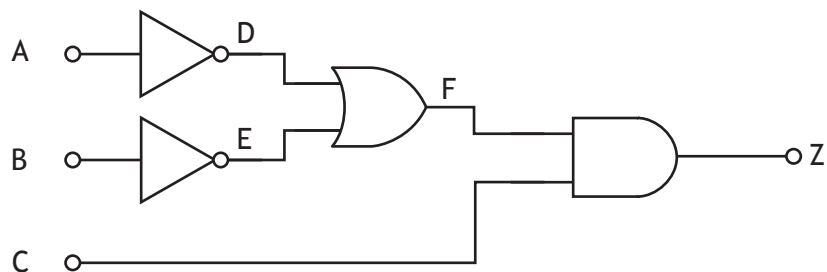
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14. (continued)

The temperature inside the trailer is monitored.

Part of the logic circuit used in the temperature monitoring system is shown below.



(c) Complete the truth table for the logic circuit shown.

3

A	B	C	D	E	F	Z
0	0	0	1			
0	0	1	1			
0	1	0	1			
0	1	1	1			
1	0	0	0			
1	0	1	0			
1	1	0	0			
1	1	1	0			

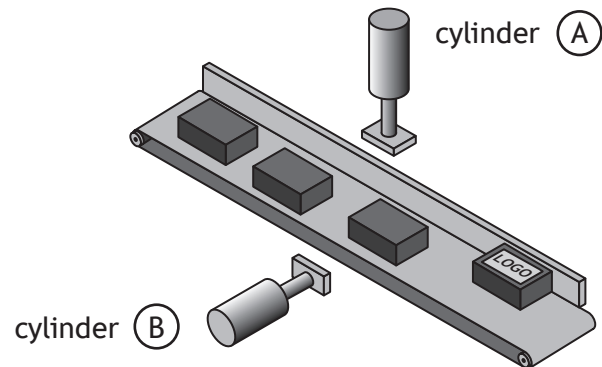


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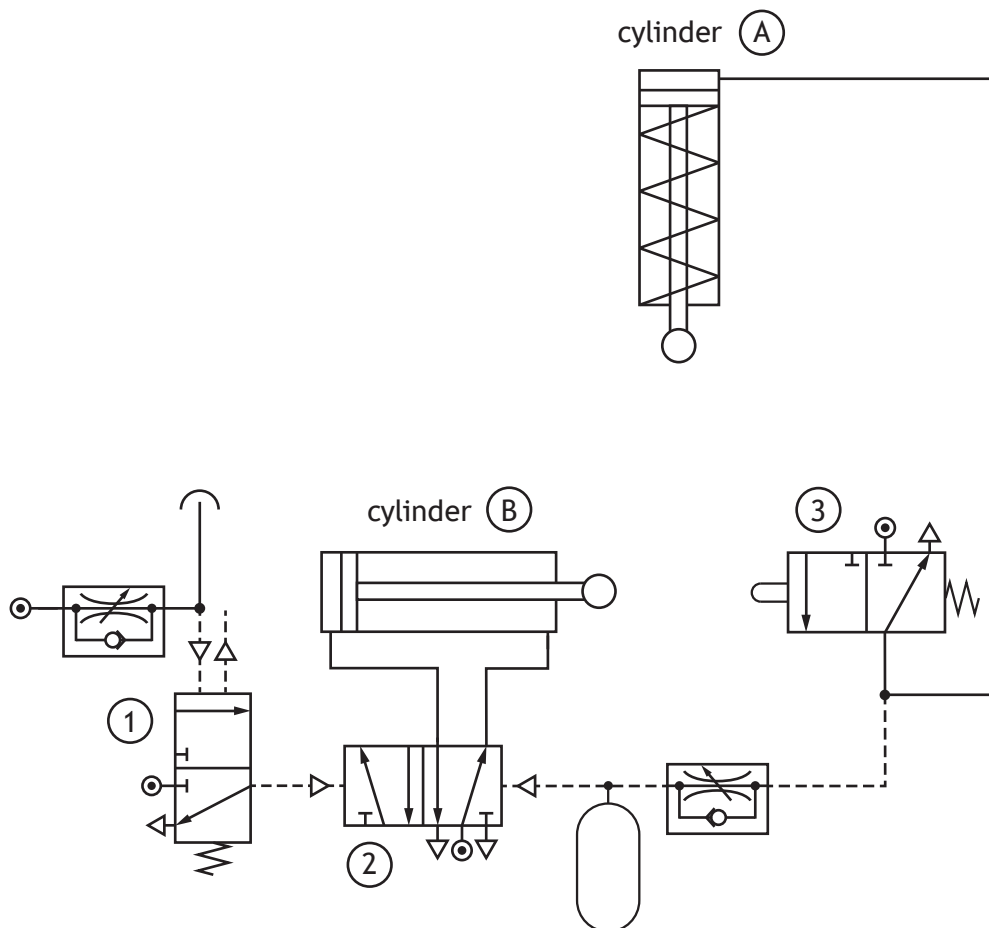
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15. Pneumatics is used to stamp a logo onto products.



A diagram for a pneumatic circuit that is used to stamp the logo onto the products is shown below.



15. (continued)

- 5

When the air bleed is covered valve (1) is actuated . . . _____

[illegible]

Cylinder (B) requires an outstroking force of 18 N.

- 3

Case No.	Case Name	Case Type	Case Status	Case Date	Case Location	Case Description	Case Details	Case Notes	Case Comments	Case Actions
1	John Doe	Case 1	Open	2023-01-01	New York	Case 1 Description	Case 1 Details	Case 1 Notes	Case 1 Comments	Case 1 Actions
2	Jane Smith	Case 2	Closed	2023-01-02	California	Case 2 Description	Case 2 Details	Case 2 Notes	Case 2 Comments	Case 2 Actions
3	Bob Johnson	Case 3	Pending	2023-01-03	Texas	Case 3 Description	Case 3 Details	Case 3 Notes	Case 3 Comments	Case 3 Actions
4	Alice Brown	Case 4	Open	2023-01-04	Florida	Case 4 Description	Case 4 Details	Case 4 Notes	Case 4 Comments	Case 4 Actions
5	Charlie White	Case 5	Closed	2023-01-05	Illinois	Case 5 Description	Case 5 Details	Case 5 Notes	Case 5 Comments	Case 5 Actions
6	Diana Green	Case 6	Pending	2023-01-06	Ohio	Case 6 Description	Case 6 Details	Case 6 Notes	Case 6 Comments	Case 6 Actions
7	Frank Black	Case 7	Open	2023-01-07	Georgia	Case 7 Description	Case 7 Details	Case 7 Notes	Case 7 Comments	Case 7 Actions
8	Grace King	Case 8	Closed	2023-01-08	Arizona	Case 8 Description	Case 8 Details	Case 8 Notes	Case 8 Comments	Case 8 Actions
9	Henry Lee	Case 9	Pending	2023-01-09	Colorado	Case 9 Description	Case 9 Details	Case 9 Notes	Case 9 Comments	Case 9 Actions
10	Ivy Hall	Case 10	Open	2023-01-10	Connecticut	Case 10 Description	Case 10 Details	Case 10 Notes	Case 10 Comments	Case 10 Actions
11	Jack Adams	Case 11	Closed	2023-01-11	Delaware	Case 11 Description	Case 11 Details	Case 11 Notes	Case 11 Comments	Case 11 Actions
12	Karen Baker	Case 12	Pending	2023-01-12	Idaho	Case 12 Description	Case 12 Details	Case 12 Notes	Case 12 Comments	Case 12 Actions
13	Leo Clark	Case 13	Open	2023-01-13	Indiana	Case 13 Description	Case 13 Details	Case 13 Notes	Case 13 Comments	Case 13 Actions
14	Mia Evans	Case 14	Closed	2023-01-14	Iowa	Case 14 Description	Case 14 Details	Case 14 Notes	Case 14 Comments	Case 14 Actions
15	Noah Foster	Case 15	Pending	2023-01-15	Kansas	Case 15 Description	Case 15 Details	Case 15 Notes	Case 15 Comments	Case 15 Actions
16	Olivia Grant	Case 16	Open	2023-01-16	Kentucky	Case 16 Description	Case 16 Details	Case 16 Notes	Case 16 Comments	Case 16 Actions
17	Peter Harris	Case 17	Closed	2023-01-17	Louisiana	Case 17 Description	Case 17 Details	Case 17 Notes	Case 17 Comments	Case 17 Actions
18	Quinn Hill	Case 18	Pending	2023-01-18	Maine	Case 18 Description	Case 18 Details	Case 18 Notes	Case 18 Comments	Case 18 Actions
19	Rachel King	Case 19	Open	2023-01-19	Maryland	Case 19 Description	Case 19 Details	Case 19 Notes	Case 19 Comments	Case 19 Actions
20	Samuel Lee	Case 20	Closed	2023-01-20	Massachusetts	Case 20 Description	Case 20 Details	Case 20 Notes	Case 20 Comments	Case 20 Actions
21	Tina Miller	Case 21	Pending	2023-01-21	Michigan	Case 21 Description	Case 21 Details	Case 21 Notes	Case 21 Comments	Case 21 Actions
22	Uma White	Case 22	Open	2023-01-22	Minnesota	Case 22 Description	Case 22 Details	Case 22 Notes	Case 22 Comments	Case 22 Actions
23	Victor Black	Case 23	Closed	2023-01-23	Mississippi	Case 23 Description	Case 23 Details	Case 23 Notes	Case 23 Comments	Case 23 Actions
24	Wendy Green	Case 24	Pending	2023-01-24	Montana	Case 24 Description	Case 24 Details	Case 24 Notes	Case 24 Comments	Case 24 Actions
25	Xavier Hall	Case 25	Open	2023-01-25	Nebraska	Case 25 Description	Case 25 Details	Case 25 Notes	Case 25 Comments	Case 25 Actions
26	Yara Adams	Case 26	Closed	2023-01-26	Nevada	Case 26 Description	Case 26 Details	Case 26 Notes	Case 26 Comments	Case 26 Actions
27	Zoe Baker	Case 27	Pending	2023-01-27	New Hampshire	Case 27 Description	Case 27 Details	Case 27 Notes	Case 27 Comments	Case 27 Actions
28	Adam Clark	Case 28	Open	2023-01-28	New Jersey	Case 28 Description	Case 28 Details	Case 28 Notes	Case 28 Comments	Case 28 Actions
29	Bella Evans	Case 29	Closed	2023-01-29	New Mexico	Case 29 Description	Case 29 Details	Case 29 Notes	Case 29 Comments	Case 29 Actions
30	Charlie Foster	Case 30	Pending	2023-01-30	New York	Case 30 Description	Case 30 Details	Case 30 Notes	Case 30 Comments	Case 30 Actions
31	Diana Grant	Case 31	Open	2023-01-31	North Carolina	Case 31 Description	Case 31 Details	Case 31 Notes	Case 31 Comments	Case 31 Actions
32	Frank Harris	Case 32	Closed	2023-02-01	North Dakota	Case 32 Description	Case 32 Details	Case 32 Notes	Case 32 Comments	Case 32 Actions
33	Grace Hill	Case 33	Pending	2023-02-02	Ohio	Case 33 Description	Case 33 Details	Case 33 Notes	Case 33 Comments	Case 33 Actions
34	Henry King	Case 34	Open	2023-02-03	Oklahoma	Case 34 Description	Case 34 Details	Case 34 Notes	Case 34 Comments	Case 34 Actions
35	Ivy Lee	Case 35	Closed	2023-02-04	Oregon	Case 35 Description	Case 35 Details	Case 35 Notes	Case 35 Comments	Case 35 Actions
36	Jack Miller	Case 36	Pending	2023-02-05	Pennsylvania	Case 36 Description	Case 36 Details	Case 36 Notes	Case 36 Comments	Case 36 Actions
37	Karen White	Case 37	Open	2023-02-06	Rhode Island	Case 37 Description	Case 37 Details	Case 37 Notes	Case 37 Comments	Case 37 Actions
38	Leo Black	Case 38	Closed	2023-02-07	South Carolina	Case 38 Description	Case 38 Details	Case 38 Notes	Case 38 Comments	Case 38 Actions
39	Mia Green	Case 39	Pending	2023-02-08	South Dakota	Case 39 Description				



15. (continued)

The piston in the double-acting cylinder is required to outstroke **slowly** and **smoothly**. The speed of the instroke should not be changed.

(c) Describe how this would be achieved.

2

(d) State the type of **motion** a pneumatic piston produces when it outstrokes and then instrokes.

1

[END OF QUESTION PAPER]



ADDITIONAL SPACE FOR ANSWERS



ADDITIONAL SPACE FOR ANSWERS

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